

Marketing ROI Analysis Using Simple Linear Regression

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import pandas as pd
```

Load the Dataset

The dataset was loaded using pandas and inspected to understand its structure.

```
df = pd.read_csv("marketing_dataset.csv")
df.head()
```

	TV	Radio	Social_Media	Sales	
0	16.0	6.566231	2.907983	54.732757	
1	13.0	9.237765	2.409567	46.677897	
2	41.0	15.886446	2.913410	150.177829	
3	83.0	30.020028	6.922304	298.246340	
4	15.0	8.437408	1.405998	56.594181	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   TV              4562 non-null   float64
 1   Radio           4568 non-null   float64
 2   Social_Media    4566 non-null   float64
 3   Sales           4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

Descriptive Statistics

Descriptive statistics were generated to summarize the key characteristics of the dataset. The output provides information on the count, mean, standard deviation, minimum, maximum, and quartile values for each variable. These statistics help in understanding the distribution and scale of the advertising expenditure and sales variables before conducting regression analysis.

```
df.describe()
```

	TV	Radio	Social_Media	Sales	
count	4562.000000	4568.000000	4566.000000	4566.000000	
mean	54.066857	18.160356	3.323956	192.466602	
std	26.125054	9.676958	2.212670	93.133092	
min	10.000000	0.000684	0.000031	31.199409	
25%	32.000000	10.525957	1.527849	112.322882	
50%	53.000000	17.859513	3.055565	189.231172	
75%	77.000000	25.649730	4.807558	272.507922	
max	100.000000	48.871161	13.981662	364.079751	

```
df = df.dropna()
```

Missing values were identified in all variables. To ensure accurate regression analysis, rows containing missing values were removed using the dropna() function. After cleaning, the dataset contained no missing values.

```
df.isnull().sum()
```

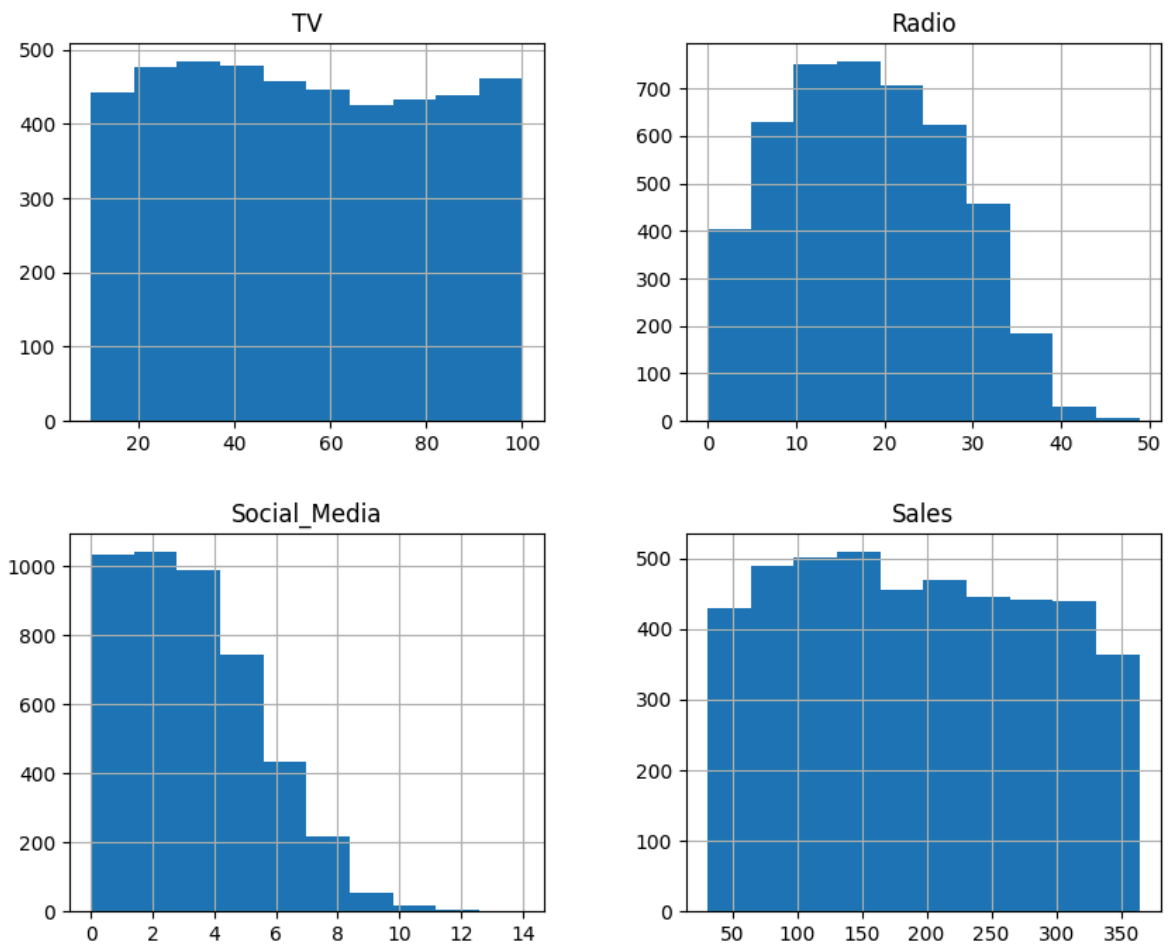
	0
TV	0
Radio	0
Social_Media	0
Sales	0

dtype: int64

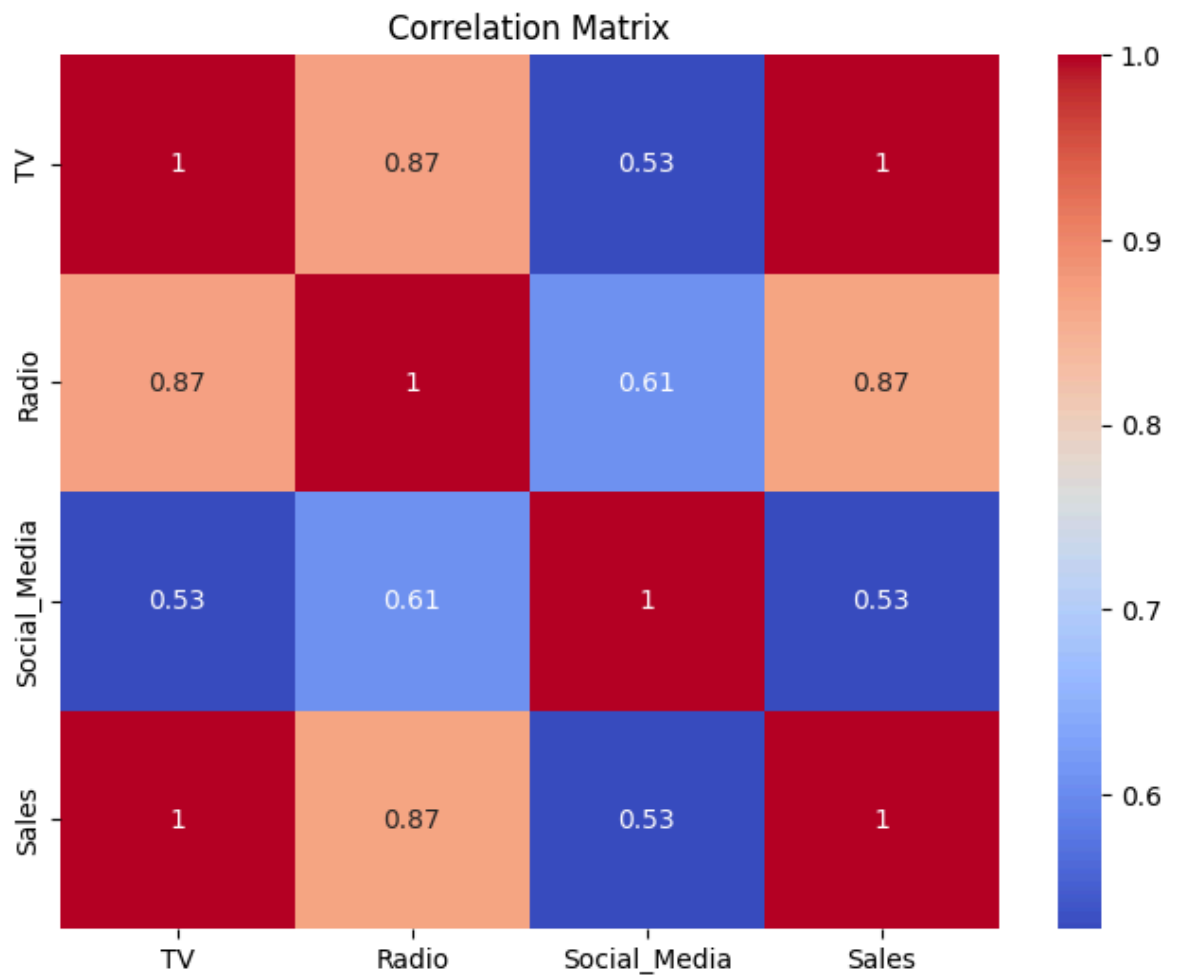
```
df.shape
```

```
(4546, 4)
```

```
df.hist(figsize=(10, 8))  
plt.show()
```



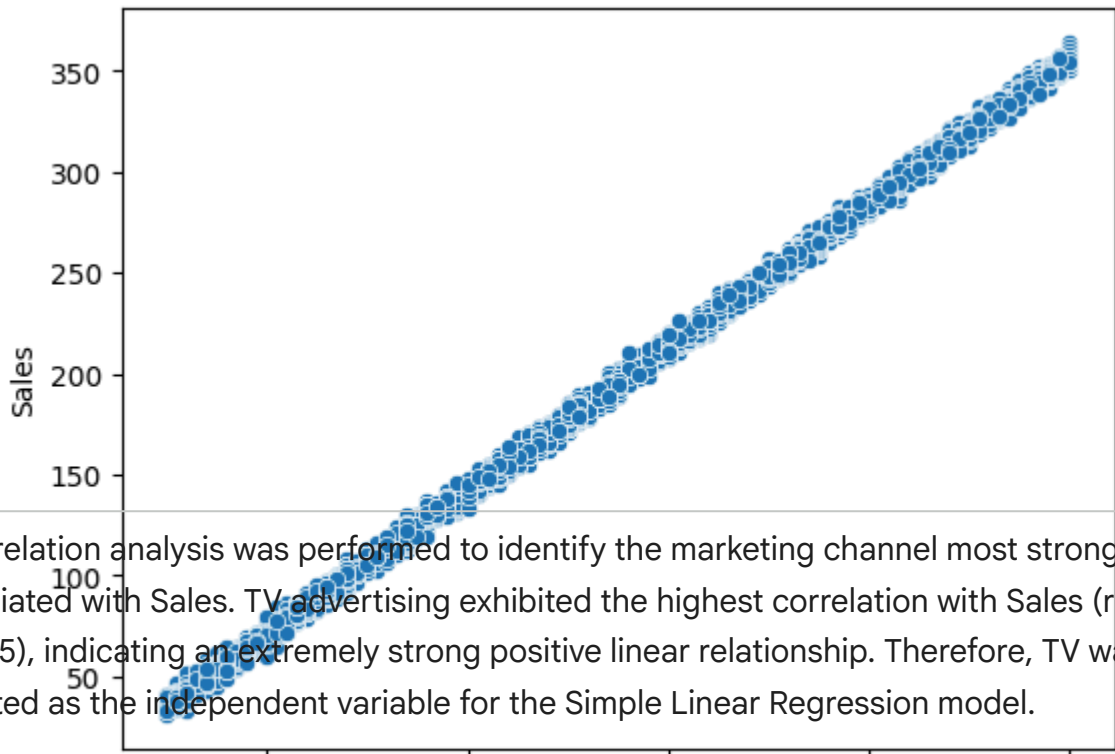
```
plt.figure(figsize=(8, 6))  
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')  
plt.title("Correlation Matrix")  
plt.show()
```



```
sns.scatterplot(x='TV', y='Sales', data=df)
plt.show()

sns.scatterplot(x='Radio', y='Sales', data=df)
plt.show()

sns.scatterplot(x='Social_Media', y='Sales', data=df)
plt.show()
```

A correlation analysis was performed to identify the marketing channel most strongly associated with Sales. TV advertising exhibited the highest correlation with Sales ($r = 0.9995$), indicating an extremely strong positive linear relationship. Therefore, TV was selected as the independent variable for the Simple Linear Regression model.

```
df.corr()
```

	TV	Radio	Social_Media	Sales
TV	1.000000	0.869158	0.527687	0.999497
Radio	0.869158	1.000000	0.606338	0.868638
Social_Media	0.527687	0.606338	1.000000	0.527446
Sales	0.999497	0.868638	0.527446	1.000000

Simple Linear Regression Model

An OLS regression model was developed using TV advertising as the predictor variable and Sales as the response variable.

Model Performance

The regression model achieved an R-squared value of 0.999. This indicates that approximately 99.9% of the variation in Sales can be explained by TV advertising expenditure. The result suggests an exceptionally strong relationship between TV advertising and Sales.

Coefficient Interpretation

The coefficient for TV advertising is 3.5615. This indicates that for every additional unit invested in TV advertising, Sales are expected to increase by approximately 3.56 units on average. The positive coefficient confirms that increased TV advertising expenditure is associated with higher Sales.

Statistical Significance

The p-value associated with TV advertising is less than 0.05, indicating that TV advertising is a statistically significant predictor of Sales. Therefore, there is strong statistical evidence that TV advertising expenditure influences Sales.

```
import statsmodels.api as sm

X = df['TV']
y = df['Sales']

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())

#
```

OLS Regression Results

```
=====
Dep. Variable:          Sales      R-squared:                0.99
Model:                  OLS       Adj. R-squared:           0.99
Method:                 Least Squares   F-statistic:             4.517e+0
Date:                  Sat, 27 Jun 2026   Prob (F-statistic):       0.0
Time:                  10:10:17    Log-Likelihood:          -11366
No. Observations:      4546        AIC:                    2.274e+0
Df Residuals:          4544        BIC:                    2.275e+0
Df Model:              1
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975
const	-0.1325	0.101	-1.317	0.188	-0.330	0.06
TV	3.5615	0.002	2125.272	0.000	3.558	3.56

```
=====
Omnibus:                0.052    Durbin-Watson:           1.99
Prob(Omnibus):          0.974    Jarque-Bera (JB):        0.03
Skew:                   -0.001    Prob(JB):                0.98
Kurtosis:               3.012    Cond. No.                138
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correct

```
predictions = model.predict(X)
residuals = y - predictions
```

✓ Residual Analysis

A Residuals vs Fitted plot was generated to evaluate the assumptions of linearity and homoscedasticity. The residuals appear to be randomly scattered around the zero line without any obvious systematic pattern or curvature. Additionally, the spread of residuals remains relatively constant across the range of fitted values. These observations suggest that the assumptions of linearity and homoscedasticity are reasonably satisfied, indicating that the linear regression model is appropriate for the data.

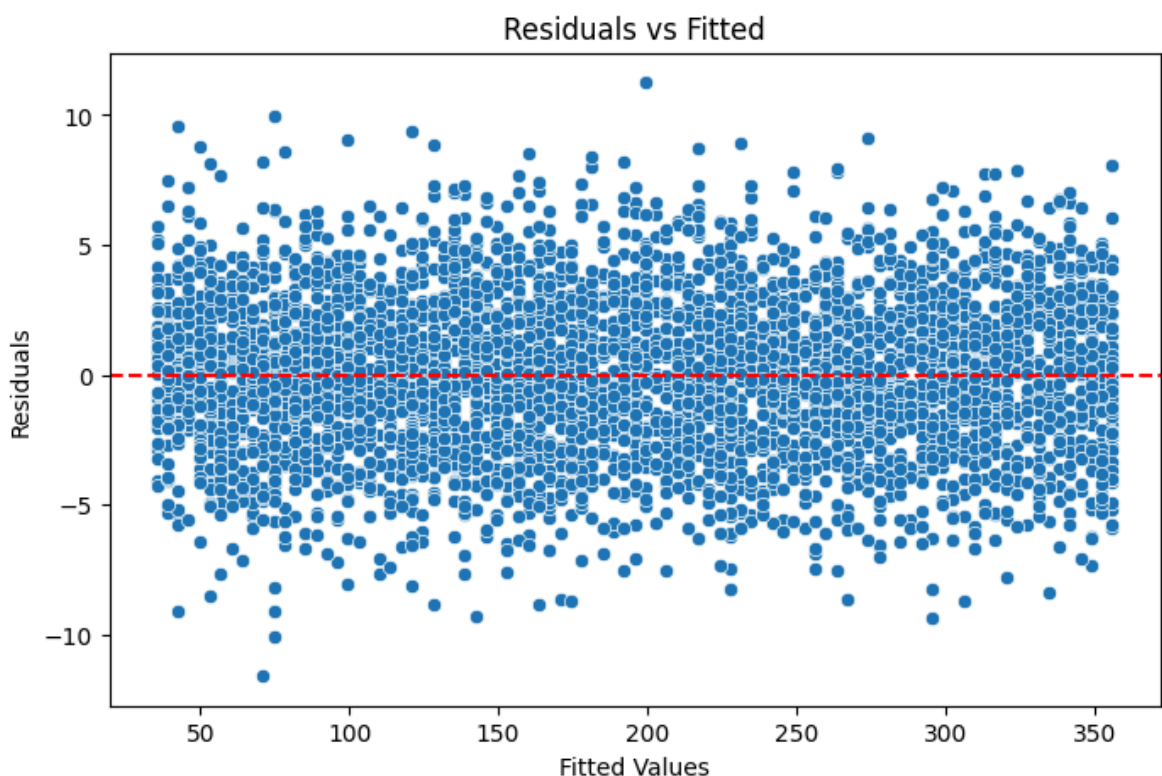
```
plt.figure(figsize=(8,5))

sns.scatterplot(x=predictions, y=residuals)

plt.axhline(y=0, color='red', linestyle='--')

plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.title("Residuals vs Fitted")

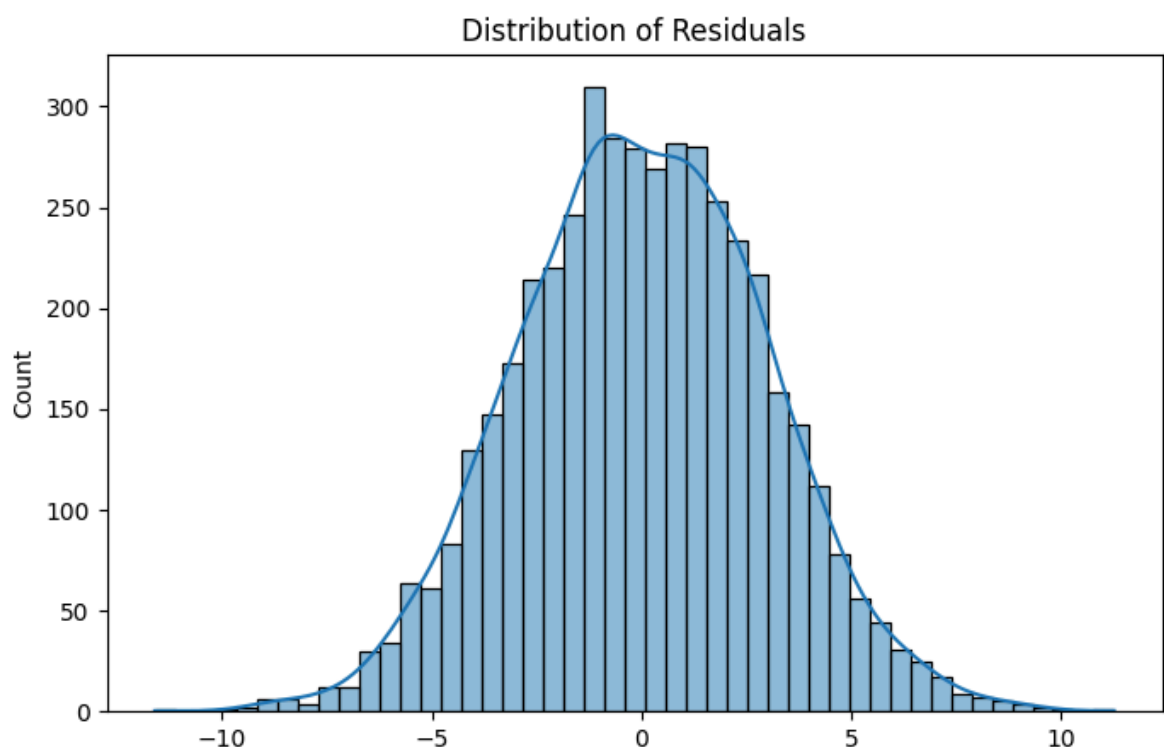
plt.show()
```



✓ Normality Assessment

A histogram of the residuals was generated to assess the normality assumption of the regression model. The residuals exhibit an approximately bell-shaped distribution centered around zero. The distribution appears reasonably symmetric, with no evidence of severe skewness or multiple peaks. Although minor deviations from perfect normality are present, they are not substantial enough to raise concerns about the validity of the model. Therefore, the normality assumption is considered to be reasonably satisfied.

```
plt.figure(figsize=(8,5))  
  
sns.histplot(residuals, kde=True)  
  
plt.title("Distribution of Residuals")  
  
plt.show()
```



✓ Q-Q Plot Analysis

A Q-Q plot was generated to further assess the normality of the residuals. Most of the residuals follow the reference line reasonably closely, indicating that the residual distribution approximates a normal distribution. While minor deviations are visible at the extreme ends of the distribution, these departures are not substantial. Combined

with the histogram of residuals and the normality statistics reported in the regression output, the results suggest that the normality assumption is reasonably satisfied.

```
sm.qqplot(residuals, line='45')  
  
plt.title("Q-Q Plot")  
  
plt.show()
```

